

Gaze-based Anxiety Sensitive Virtual Social Communication Platform for Individuals with Autism

Pradeep Raj Krishnappa Babu*

Electrical and Computer Engineering, Pratt School of Engineering, Duke University, USA; Cognitive Science, Indian Institute of Technology Gandhinagar, India
pradeep.rj1@gmail.com

Uttama Lahiri

Electrical Engineering, Indian Institute of Technology Gandhinagar, India
uttamalahiri@iitgn.ac.in

ABSTRACT

Difficulties in social communication along with impairments in understanding other's emotions and exhibiting atypical gaze behavior while avoiding social cues are common concerns in individuals with autism spectrum disorder (ASD). Such deficits are reported to be related with social anxiety. Social anxiety often deters the process of skill learning. Given the dramatic rise in the prevalence of ASD and limited availability of trained therapists, getting access to technology-assisted platforms that can estimate one's anxiety and autonomously vary the task challenges for effective skill learning is critical. The potential of gaze-related indices to serve as biomarkers to one's anxiety, and advancement in computing technologies allowing real-time access to gaze-related social signals, we have designed a Virtual Reality (VR)-based anxiety-sensitive social communication task platform. This features a Rule Generator that is used to offer tasks based on the composite effect of task performance and estimated anxiety from one's gaze in an individualized manner. Results of a study with 10 individuals with ASD showed that the anxiety-sensitive system showed promise in eliciting improved task performance and increased looking towards the face region of a communicator displaying emotional expression in comparison to the currently existing system that is blind to one's anxiety. This paves the way towards design of a complementary tool for the therapists.

CCS CONCEPTS

• **Human-centered computing** → Human computer interaction (HCI); Empirical studies in HCI; Human computer interaction (HCI); Interaction paradigms; Virtual reality.

KEYWORDS

Affective Computing, Anxiety, Autism, Looking Pattern, Virtual Reality, Flow Theory

ACM Reference Format:

Pradeep Raj Krishnappa Babu* and Uttama Lahiri. 2022. Gaze-based Anxiety Sensitive Virtual Social Communication Platform for Individuals with

Autism. In *CHI Conference on Human Factors in Computing Systems Extended Abstracts (CHI '22 Extended Abstracts)*, April 29–May 05, 2022, New Orleans, LA, USA. ACM, New York, NY, USA, 7 pages. <https://doi.org/10.1145/3491101.3519855>

1 INTRODUCTION

Autism spectrum disorder is characterized by difficulties in social communication that can be related to idiosyncratic gaze behavior causing them to miss the social cues e.g., emotional expressions [2]. Looking pattern is an important ingredient in establishing fluid interpersonal social communication. However, social anxiety being a predominant factor among individuals with ASD [21], they often exhibit atypical gaze behavior [9], resulting in challenges to recognize a partner's facial emotional expressions [30] and establish fluid social interaction [21, 30]. Social communication skill training for individuals with ASD while harnessing on other's emotional expressions are critical [33]. This can be addressed by experienced therapists while picking up one's anxiety and adapt the intervention paradigm for effective individualization.

With dramatic rise in the prevalence of ASD, it is difficult to avail one-to-one floor-time therapy from experienced therapists due to limited availability of these trained therapists [14]. Therefore, getting access to technology-assisted platforms to offer skill training while automatically estimating and adapting to one's anxiety is essential. Sensing the social anxiety of individuals with ASD becomes challenging since they cannot show explicit expression of affective states [2]. Thus, instead of picking up one's vocal intonation or facial expressions, tapping implicit measures, such as gaze and physiological measures [1, 16, 18, 31, 32] to estimate one's social anxiety is a viable alternative.

Acquiring and deciphering the gaze-related social signals in real-time is possible given the recent advancement in computing technologies [1, 7, 16, 17]. This can help open avenues to develop assistive human-computer interaction platforms [3]. Given that the individuals with ASD have an affinity towards computers [5], designing human-computer interaction platforms that can use rich computer graphics and virtual reality (VR) to project social communication scenarios while estimating the social anxiety in real-time can be a promising avenue [11, 13]. Added to offering audio-visual interface, the versatility of the VR platform [9] can be utilized to design variety of social tasks that can be presented while autonomously adapting to one's social anxiety in an individualized manner [26].

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than ACM must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

CHI '22 Extended Abstracts, April 29–May 05, 2022, New Orleans, LA, USA

© 2022 Association for Computing Machinery.

ACM ISBN 978-1-4503-9156-6/22/04...\$15.00

<https://doi.org/10.1145/3491101.3519855>

1.1 Related works

Usage of dynamic stimuli with video-based technique has emerged to be promising in teaching social skills for individuals with autism [22]. Video-based approach uses human actors (characters) to narrate social stories while displaying context-relevant emotional expressions [22]. However, video-based techniques are not anxiety-sensitive and does not feature real-time modification of the instructions or the emotional expressions shown by the characters, limiting the potential of adapting the stimuli/task challenges based on one's anxiety, critical for effective learning [32].

Given the importance of real-time adaptation, VR can be used as an avenue to modify a simulated context in real-time and in a controlled manner [19, 32]. The presently existing VR-based platforms operate in an pseudo closed-loop fashion that is adaptive to only one's performance accessed from one's performance score [20] that might work well for designing game engines, but not necessarily for skill learning for individuals with ASD. Combining both the performance and anxiety measures for effective individualization while developing an adaptive platform is crucial in offering a rich learning experience [20, 32]. This emphasizes the need of incorporating implicit modalities for sensing one's affective (anxiety) state in a VR-based setting [17]. Since acquiring gaze data is easier and have faster response time than the peripheral physiological signals [17, 24], and literature indicating that gaze data can give richer information in emotion-related tasks than one's peripheral physiology [25, 29], researchers had been focusing on tapping gaze-related indices. Looking pattern quantified in terms of Fixation Duration (FD) [1] can give an estimate on atypical gaze behavior in ASD due to social anxiety [7, 9] and eye physiology, e.g., Pupil Diameter (PD) and Blink Rate (BR) being involuntary measures can help in estimating one's anxiety while exposed to social tasks [16].

1.2 Current work and objectives

Since the individuals with ASD have affinity towards computers, we utilize the benefits of VR [23] as a tool to develop our Virtual Social communication Platform (VISP) for children with ASD that is sensitive to one's social anxiety (estimated from one's gaze) and task performance. The objectives of the current work were to (1) design PAS-VISP (i.e., Performance and Anxiety Sensitive-VISP), (2) design PS-VISP (Performance Sensitive-VISP) to replicate a pseudo closed-loop platform, and (3) study the effectiveness of the PAS-VISP compared to PS-VISP manifested in terms of one's skill learning and looking pattern while exposing some participants with autism to both the platforms. We hypothesized that PAS-VISP would emerge promising than PS-VISP in terms of effectiveness in social skill training.

2 DESIGN OF THE VIRTUAL REALITY BASED PLATFORMS

Facial emotional expressions and the emotion-recognition being the important aspect of social communication in this research work, we have developed the VR-based tasks with avatars displaying context-relevant dynamic facial emotional expressions while standing in a context-relevant VR-based background. The avatars narrated short social stories [6], demonstrating their experience about a social situation.

2.1 Choice of avatars and emotional expressions

The avatar faces were programmed to display three different emotional expressions, namely, Happy, Angry and Neutral [18]. A survey on these designed faces was conducted in a pilot study. The survey was to ensure and validate that the avatar's facial emotional expression was interpreted correctly by the viewers and appeared Indian, to comply with the population of interest. A total of 24 heads comprising of 4 male and 4 female faces displaying 3 emotions were designed and used in the survey. Based on the survey, 3 male and 3 female faces were chosen based on the specific valence and arousal level (see [18] for more details). Also, for a naturalistic conversation the avatars could exhibit a balanced mixture of straight and averted gaze [15], speak in lip-synchronized manner, blink their eyes and walk in the VR environment.

2.2 VR-based tasks

The VR-based tasks were developed using Vizard software (from Worldvz Inc.). The VR environments for the tasks belonged to three categories, e.g., *i) home based closed space* (less crowded, e.g., get-together party at home), *ii) non-home based closed space* (partially crowded, e.g., classroom) and *iii) open space* (crowded, e.g., football match stadium) similar to [16]. In each task, an avatar narrated a social story in a form of his/her own experience (~1 min long). During this process, the avatar exposed the participants to some of the basic elements of social skills, such as *being courteous*, *making friends*, *participation in a social task*, *following discipline* and *social etiquettes* [34]. For example, in a christmas party event (Figure 1(a)) an avatar walked into a decorated house, wished 'hello' to the participant (i.e., *being courteous*). Then she went on sharing her experience of making new friends who came to the party (i.e., *making friends*), enjoying and learning how to bake a cake with her friends (i.e., *participation*), serving cake to her friends before taking one to herself in the plate (i.e., *following discipline* and *social etiquettes*). Based on the context of the story, the avatar demonstrated a context-relevant facial emotional expression, which was Happy for this example.

Subsequent to the story narration, the participants were expected to respond to a set of five questions (each with three options) using a menu-driven interface (Figure 1 (b)). The participants could respond to the avatar using a number keypad. The tasks were categorized into three challenge levels based on the type of questions (such as context-related, emotion-recognition, etc.) being asked to the participants, similar to that in [16]. Greater weightage in scoring was allotted to questions on emotion-recognition, requiring one to decipher the avatar's facial emotional expression during story narration in the tasks belonging to increased challenging level.

2.3 Gaze data acquisition

A ViewPoint eye tracker, a light-weight wearable goggles from Arrington Research, Inc. was integrated with VR software to extract the participant's gaze data. We acquired one's raw gaze data, e.g., FD, PD, pupil major axis and 2D gaze coordinates. These raw data were subsequently used to process and extract one's gaze-related behavioral and physiological indices during each task (similar to [16]). The visual stimulus was segmented into two Regions of Interest

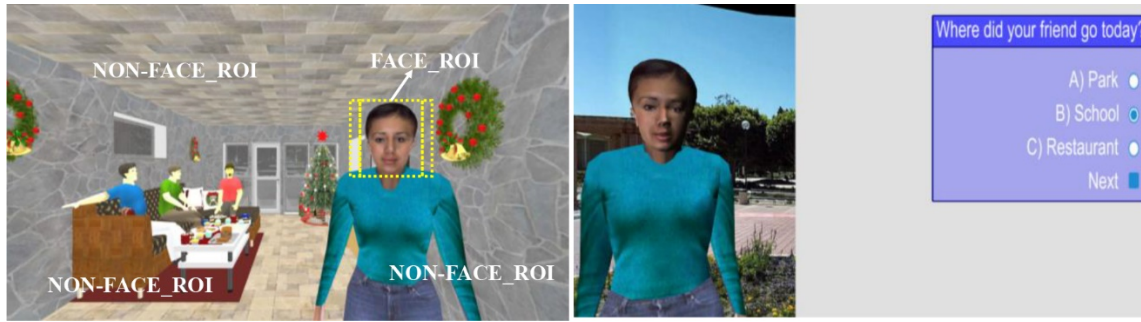


Figure 1: VR-based task module with (a) various Regions of Interest (ROI) used for eye tracking and (b) participant-avatar interaction

(ROIs) e.g., avatar’s face (FACE_ROI) and the rest (NON-FACE_ROI) (Figure 1(a)). The behavioral indices corresponding to the two ROIs were i) Total Fixation Duration (FD_{FACE} and $FD_{NON-FACE}$). The physiological indices were, i) Average Pupillary Diameter (PD_{AVG}) and ii) Mean Blink Rate (BR_{MEAN}).

2.4 Adaptive task switching

To offer tasks belonging to different challenge levels, namely Task Type 1, 2 and 3 (TT1, TT2, TT3), we developed state machine-based task switching rationale. Being a traditional pseudo closed-loop platform, the PS-VISP used the task performance (i.e., learning metrics) to offer tasks belonging to different challenge levels. If the performance was *Adequate* (i.e., condition (C1): %Performance >70%; Figure 2(a)), one was offered a task of higher challenge and vice versa (condition (C2)).

In contrast, the PAS-VISP used a *Rule Generator* that switched tasks based on the composite effect of task performance and anxiety predicted from gaze-related indices (Figure 3). The *Rule Generator* for PAS-VISP was designed with the concept of *Flow Theory* [10], inspired from existing computer-mediated environment [12, 27]. According to the *Flow Theory*, an individual performs a given task with *Flow* using his full capacity, provided he/she manages to maintain a balance of his/her skill with the challenge that he/she experiences during task execution. If the challenge surpasses the skill level, one can suffer from *panic* (due to anxiety). On the other hand, if the skill level is beyond the current challenge, one might feel relaxed and subsequently *bored* if the condition prevails. Based on task performance (*‘Adequate’* or *‘Inadequate’*) and estimated anxiety (*‘High’* or *‘Low’*), one’s interaction was labeled to indicate (i) *Boredom*, (ii) *Flow* and (iii) *Panic/Anxious* (Table 1).

The *Gaze-Indices extractor* in the *Rule Generator* (Figure 3) computed the change in the gaze-related physiological indices during a task i from that of its immediately preceding task ($i-1$) for the PD and BR as δPD and δBR , respectively. For the behavioral indices, the percentage of the time the participant fixated towards the FACE_ROI ($\%FD_{FACE}$) and the ratio of $FD_{NON-FACE}$ to FD_{FACE} (i.e., NFFR normalized to 0-1) during the story narration was estimated. Changes (i.e., increase in δPD and δBR , and decrease in $\%FD_{FACE}$ and NFFR) in each of the gaze-related indices was categorized into 4 discrete levels of anxiety, e.g., 1, 2, 3 and 4 (Table 2). To decide the numerical threshold for the anxiety levels, we have used our

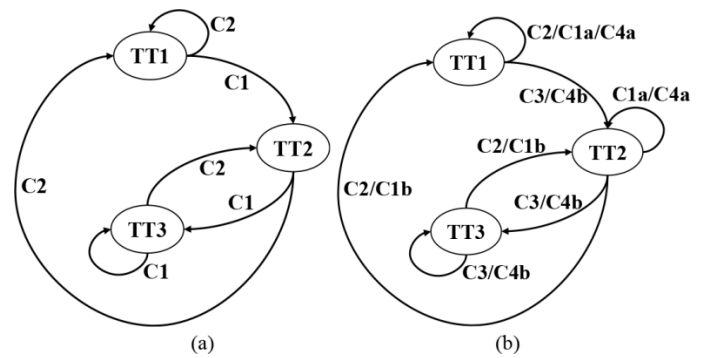


Figure 2: State machine representations

findings from previous studies of individuals with ASD exposed to similar gaze-sensitive VR-based platform [16] that were validated with the ratings (on the estimates of anxiety) given by the therapist involved in the study.

If the participant was in *Flow* state, he/she was expected to perform *Adequately* with *Low* anxiety (Condition C3: Table 1), causing the *Rule Generator* to offer him a task of increased challenge; state machine (Figure 2(b)) moving the participant from TT1-TT2-TT3. In contrast, if one’s anxiety was *High* and his/her performance was *Inadequate* (≤ 70) (i.e., Condition C2 (Table 2)), he/she was considered to be in the *Panic* state. Then the state machine (Figure 2(b)) moved the participant to a task of lower challenge level i.e., TT3-TT2-TT1 with the hope of bringing him/her back to the *Flow* state. Again, there are two other critical conditions, e.g., C1 and C4 (Table 2). Under C1, though the participant could score *Adequate* performance, he/she might show *High* anxiety that might be because he/she has been pressing himself/herself too hard to score well and in turn may cause him/her to be in the *Panic* state. Offering him/her a task of increased challenge at this state might adversely affect his/her performance. In turn, we offered the participant, a task in the same challenge level (i.e., same Task Type; condition C1a) with the hope that his/her anxiety will reduce. If the *Panic* state prevailed in the subsequent task (condition C1b) then a task of lower challenge was offered with the hope that he/she move to the *Flow* state. Again, same as C1, in C4, we offered the participant with one more chance in the same challenge level (condition C4a;

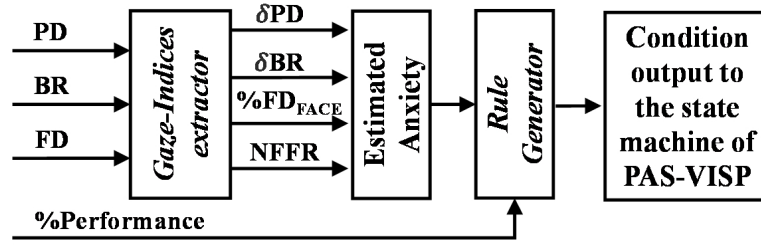


Figure 3: Architecture of Rule Generator for PAS-VISP

Table 1: Conditions for task switching in PAS-VISP

Condition	Anxiety	%Performance	State	Action
C1a/b	High	Adequate	Panic	→ / ↓
C2	High	Inadequate	Panic	↓
C3	Low	Adequate	Flow	↑
C4a/b	Low	Inadequate	Boredom	→ / ↑

Table 2) considering the participant was in the *Boredom* state. If *Boredom* prevailed, it might imply that the task (belonging to the current Task Type) was not challenging for the participant. Under that condition (C4b), a task of higher challenge was offered.

3 METHODS

3.1 Participants

We recruited 10 individuals with high-functioning ASD (5 males and 5 females) who were assessed with standard clinical scales, such as Social Responsive Scale (SRS) [8] and Social Communication Questionnaire (SCQ) [28]. All the participants were above the clinical cut-off (SRS = 60 and SCQ = 15) in either/both the scores. All the participants were enrolled in special needs school, and they were comfortable communicating in English.

3.2 Experimental setup, study protocol and ethics

The experimental setup comprised of a ViewPoint eye tracker, task computer with a keypad. A commitment of ~45 minutes each day was expected from each volunteer in the study. We exposed the participants to both PS-VISP and PAS-VISP on two different days with an intermediate refresh period between the exposures. We counterbalanced the exposure to the platforms by offering PS-VISP for half of the participants during the first exposure followed by

PAS-VISP and vice-versa for the rest of the participants. To nullify any practice effect, the story used for narration, and the questions asked were chosen to be different for both the platforms. A therapist accompanied the participant during the study. The participant was introduced to the experimental setup followed by explaining the steps to interact with the VR-based tasks. All the participants were informed that they were free to withdraw from the study at any time if they felt uncomfortable. The participants signed the informed consent. Also, a verbal consent was taken from the caregivers. Once the participant was ready, the experimenter helped the participant to wear the eye tracker and executed 16-point eye tracker calibration procedure and started the VR-based task interaction. The study protocol was reviewed and approved by the institutional ethics committee.

3.3 Statistical analysis

We have used dependent sample t-test to evaluate statistically differential implications of PS-VISP and PAS-VISP on the participants' (i) task performance and (ii) looking behavior, along with effect size (Cohen's d) [4].

4 RESULTS AND DISCUSSION

Here we analyzed the participants' task performance and looking pattern in terms of fixation duration towards the task stimuli. These measures were compared between the PS-VISP and PAS-VISP while the participants were exposed to tasks of varying challenges (TT1, TT2 and TT3) and avatar's varying emotional expressions (Happy, Angry and Neutral).

Table 2: Anxiety predictor chart using gaze-related indices

Anxiety Level	δPD	δBR	$\%FD_{FACE}$	NFFR (X, say)
1	$0 \leq \delta PD < 1.015$	$0 \leq \delta BR < 1.015$	$\geq 70\%$	$X > (30/70)$
2	$1.015 \leq \delta PD < 1.065$	$1.015 \leq \delta BR < 1.075$	$< 70\% \ \& \ \geq 50\%$	$(30/70) > X \ \& \ X \leq (40/60)$
3	$1.065 \leq \delta PD < 1.125$	$1.075 \leq \delta BR < 1.265$	$< 50\% \ \& \ \geq 30\%$	$(40/60) > X \ \& \ X \leq 50/50$
4	$1.125 \leq \delta PD$	$1.265 \leq \delta BR$	$< 30\%$	$X > (50/50)$

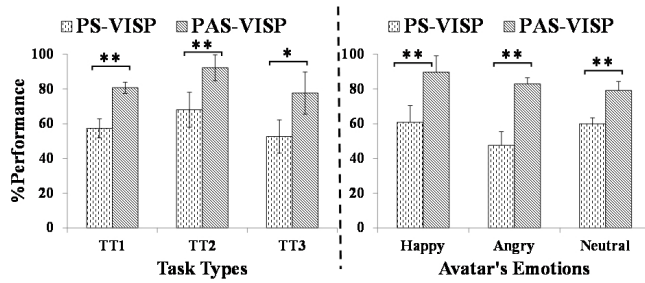


Figure 4: Comparison of %Performance (Symbols: **→ $p<.01$ /*→ $p<.05$; The mean difference had a very large effect size ($d>1.4$) for all the cases.)

4.1 Comparative analysis of participants' task performance while interacting with PS-VISP and PAS-VISP

While participants responded to the questions asked by the avatars, their %Performance was computed. The group average %Performance was significantly higher for PAS-VISP than PS-VISP, irrespective of the Task Type and emotional expressions (Figure 4). Considering only the Task Types, the performance was the maximum for TT2 that might be because the participants were already habituated to the task environment post exposure to tasks belonging to TT1. Again, the %Performance was minimum for TT3 that might be because the participants found it as the most difficult. In addition, we found that the participants could score *Adequate* performance in PAS-VISP irrespective of the Task Type, while it was *Inadequate* during PS-VISP with the mean difference in performance between the two platforms being $\Delta TT1=40.7\%$, $\Delta TT2=35.6\%$ and $\Delta TT3=47.4\%$. While analyzing the performance in response to different emotional expressions, it was found to be the maximum for Happy, though still *Inadequate* for PS-VISP irrespective of the emotional expressions, unlike PAS-VISP. The difference between the two platforms in response to avatar's emotions were: $\Delta Happy=47.5\%$, $\Delta Angry=74.4\%$ and $\Delta Neutral=31.8\%$.

4.2 Comparative analysis of participants' looking behavior while interacting with PS-VISP and PAS-VISP

The Figure 5 shows that the group average %FD_{FACE} for PAS-VISP was significantly higher than PS-VISP, irrespective of the Task Type and emotional expressions. With regard to the Task Type, for PS-VISP, we see that the trend of the variation in %FD_{FACE} was similar to that of their task performance (Figure 4). In contrast, for PAS-VISP, we can see that the %FD_{FACE} was nearly the same for TT1 and TT2, while there was a dip in TT3. The mean percentage change was significantly ($p<.05$ and $d>1.4$) different between the two platforms: $\Delta TT1=96.1\%$, $\Delta TT2=52.5\%$, $\Delta TT3=54.6\%$. Again, this observation is similar to the variation in their task performance. With regard to the emotional expressions, for PS-VISP, the %FD_{FACE} was the least for Angry expression, as expected that in turn was reflected in their task performance. In contrast, for PAS-VISP, the %FD_{FACE} was nearly the same for all the emotions causing them to

achieve *Adequate* performance across all the emotional expressions. The mean percentage change was significantly different ($p<.05$ and $d>1.5$) between the platforms: $\Delta Happy=48.4\%$, $\Delta Angry=113.2\%$ and $\Delta Neutral=36.0\%$.

In addition, while comparing the %FD_{NON-FACE} across different Task Types, the PAS-VISP was not significantly different from PS-VISP (though the effect size was still large; $d>.8$) with the mean difference between the two platforms being $\Delta TT1=15.9\%$, $\Delta TT2=15.4\%$, $\Delta TT3=62.6\%$. But while comparing the %FD_{NON-FACE} across varying emotional expressions, the PAS-VISP had significantly higher value with large effect size ($p<.05$, $d>1.4$) than PS-VISP with the mean difference between the two platforms being $\Delta Happy=38.0\%$, $\Delta Angry=25.3\%$ and $\Delta Neutral=27.5\%$ (Figure 5). Another interesting observation was that the cumulative sum of %FD_{FACE} and %FD_{NON-FACE} was significantly higher with large effect size ($p<.05$ and $d>0.8$) for PAS-VISP than PS-VISP across Task Types and emotional expressions. Such an observation might indicate that the participants attended more towards the visual stimuli when offered with PAS-VISP than PS-VISP.

5 CONCLUSION

Here, we have presented the design of Virtual Reality-based Social communication task Platform, PAS-VISP that is adaptive to one's performance and anxiety state (predicted from gaze-related indices). In addition, we conducted a study with ten participants with ASD while comparing PAS-VISP with traditional VR-based platform that is adaptive to only one's task performance. For this, we also designed PS-VISP (sensitive to only task performance) emulating the currently existing traditional VR-based systems while adapting it to our study requirements. To compare between the two platforms, we analyzed the participants' task performance and looking pattern in terms of fixation duration towards the social cue such as avatar's face region.

The anxiety-sensitive system was able to elicit improved task performance than the pseudo closed-loop system that is not sensitive to one's anxiety. Our finding indicated that the participants were able to achieve improved task performance while scoring *Adequately* when interacting with PAS-VISP than that with PS-VISP. Possibly, the PAS-VISP offered one with an opportunity to operate in a *Flow* state and cope up with increased task challenge.

Participants exhibited increased fixation duration towards the avatar's face region while interacting with anxiety-sensitive system than the pseudo closed-loop system. The results demonstrated an improved looking behavior among the participants in terms of greater fixation towards the avatar's face region while interacting with PAS-VISP than PS-VISP, potentially indicating that the P-VISP offered a comfortable medium for training these individuals.

Contribution. Our research findings suggest the PAS-VISP platform that was adaptive to one's anxiety and performance showed

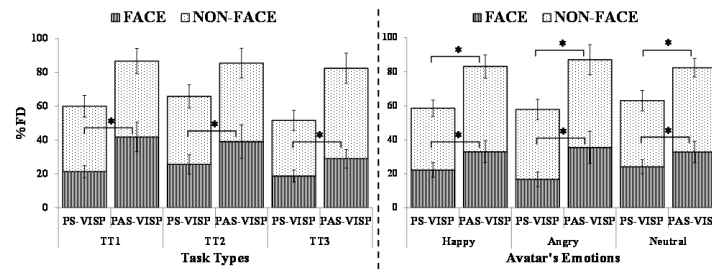


Figure 5: Comparison of Participants' %FD (Symbol: * $\rightarrow$ $p < .05$.)

promise in improving one's task performance and looking pattern than PS-VISP that was blind to one's anxiety. The anxiety-sensitive closed-loop social communication platform has a potential to serve as a complementary tool in the hands of the therapists/interventionists where one therapist can facilitate multiple individuals at the same time, partly combating with the resource constraints imposed by the growing pace in the prevalence of autism. **Limitations.** A potential limitation of our current work is the small sample size. Our present study was a proof-of-concept study that focused on understanding the benefits and potential of our anxiety-sensitive platform (PAS-VISP). However, before using this platform for intervention studies, a larger participant pool needs to be enrolled so that the inferences can be generalized across cultures, race and ethnicity.

Future work. In future, we plan to (1) increase the sample size, (2) increase the inventory of VR-based tasks that would enable us to offer these individuals with multiple exposures over prolonged duration, and (3) address the question on the optimal duration of exposure to such platforms necessary for different participants on the autism spectrum.

REFERENCES

- [1] Mariano Alcañiz, Irene Alice Chicchi-Giglioli, Lucía A. Carrasco-Ribelles, Javier Marin-Morales, Maria Eleonora Minissi, Gonzalo Teruel-García, Marian Sirera, and Luis Abad. 2022. Eye gaze as a biomarker in the recognition of autism spectrum disorder using virtual reality and machine learning: A proof of concept for diagnosis. *Autism Res.* 15, 1 (2022), 131–145.
- [2] American Psychiatric Association. 2014. Diagnostic and statistical manual of mental disorders: DSM-5. American Psychiatric Association.
- [3] André Antunes and Rui Neves Madeira. 2022. PLAY - Model-based Platform to Support Therapeutic Serious Games Design. *Procedia Comput. Sci.* 198, (2022), 211–218.
- [4] R Bakeman. 2005. Recommended effect size statistic. *Behav. Res. Methods* (2005).
- [5] Jennifer Brown and Donna Murray. 2001. Strategies for enhancing play skills for children with autism spectrum disorder. *Educ. Train. Ment. Retard. Dev. Disabil.* 36, 3 (2001), 312–317.
- [6] Louis John Camilleri, Katie Maras, and Mark Brosnan. 2022. The impact of using digitally-mediated social stories on the perceived competence and attitudes of parents and practitioners supporting children with autism. *PLoS One* 17, 1 (January 2022), e0262598.
- [7] Zhuoqing Chang, J Matias Di Martino, Rachel Aiello, Jeffrey Baker, Kimberly Carpenter, Scott Compton, Naomi Davis, Brian Eichner, Steven Espinosa, Jacqueline Flowers, Lauren Franz, Adrienne Harris, Jill Howard, Sam Perochon, Eliana M Perrin, Pradeep Raj, Krishnappa Babu, Marina Spanos, Connor Sullivan, Barbara K Walter, Scott H Kollins, Geraldine Dawson, and Guillermo Sapiro. 2021. Computational Methods to Measure Patterns of Gaze in Toddlers With Autism Spectrum Disorder. (2021), 1–10. DOI:https://doi.org/10.1001/jamapediatrics.2021.0530
- [8] John N. Constantino. 2021. Social Responsiveness Scale. In *Encyclopedia of Autism Spectrum Disorders*. DOI:https://doi.org/10.1007/978-3-319-91280-6_296
- [9] Hayley Crawford, Joanna Moss, Chris Oliver, Natasha Elliott, Giles M. Anderson, and Joseph P. McCleery. 2016. Visual preference for social stimuli in individuals with autism or neurodevelopmental disorders: An eye-tracking study. *Mol. Autism* 7, 1 (April 2016), 1–12.
- [10] Mihaly Csikszentmihalyi. 2014. Play and intrinsic rewards. In *Flow and the Foundations of Positive Psychology: The Collected Works of Mihaly Csikszentmihalyi*. 135–153. DOI:https://doi.org/10.1007/978-94-017-9088-8_10
- [11] Anders Dechsling, Stian Orm, Tamara Kalandadze, Stefan Sütterlin, Roald A. Øien, Frederick Shic, and Anders Nordahl-Hansen. 2021. Virtual and Augmented Reality in Social Skills Interventions for Individuals with Autism Spectrum Disorder: A Scoping Review. *J. Autism Dev. Disord.* (2021). DOI:https://doi.org/10.1007/s10803-021-05338-5
- [12] Christina M. Finneran and Ping Zhang. 2005. Flow in Computer-Mediated Environments: Promises and Challenges. *Commun. Assoc. Inf. Syst.* 15, 1 (2005), 4. DOI:https://doi.org/10.17705/1cais.01504
- [13] Ouriel Grynszpan, Patrice L. Weiss, Fernando Perez-Diaz, and Eynat Gal. 2014. Innovative technology-based interventions for autism spectrum disorders: A meta-analysis. *Autism* 18, 346–361. DOI:https://doi.org/10.1177/1362361313476767
- [14] Stefan G. Hofmann and Jasper A.J. Smits. 2008. Cognitive-behavioral therapy for adult anxiety disorders: A meta-analysis of randomized placebo-controlled trials. *J. Clin. Psychiatry* 69, 4 (2008), 621–632. DOI:https://doi.org/10.4088/jcp.v69n0415
- [15] Pradeep Raj Krishnappa Babu and Uttama Lahiri. 2019. Understanding the role of Proximity and Eye gaze in human-computer interaction for individuals with autism. *Journal of Ambient Intelligence and Humanized Computing*. DOI:https://doi.org/10.1007/s12652-019-01175-8
- [16] Pradeep Raj Krishnappa Babu, Poojan Oza, and Uttama Lahiri. 2018. Gaze-sensitive virtual reality based social communication platform for individuals with autism. *IEEE Trans. Affect. Comput.* 9, 4 (2018), 450–462.
- [17] Karunanithi Praveen Kumar and Perumal Sivanesan. 2021. Smart Virtual Reality-Based Gaze-Perceptive Common Communication System for Children With Autism Spectrum Disorder. *Cogn. Behav. Hum. Comput. Interact. Based Mach. Learn. Algorithm* (December 2021), 117–135.
- [18] Selvia Kuriakose and Uttama Lahiri. 2015. Understanding the Psycho-Physiological Implications of Interaction with a Virtual Reality-Based System in Adolescents with Autism: A Feasibility Study. *IEEE Trans. Neural Syst. Rehabil. Eng.* 23, 4 (2015), 665–675.
- [19] Adam Naples, Alyssa Nguyen-Phuc, Marika Coffman, Anna Kresse, Susan Faja, Raphael Bernier, and James C. McPartland. 2015. A computer-generated animated face stimulus set for psychophysiological research. *Behav. Res. Methods* 47, 2 (2015), 562–570.
- [20] Chava Peretz, Amos D. Korczyn, Evelyn Shatil, Vered Aharonson, Smadar Birnboim, and Nir Giladi. 2011. Computer-based, personalized cognitive training versus classical computer games: A randomized double-blind prospective trial of cognitive stimulation. *Neuroepidemiology* 36, 2 (2011), 91–99.
- [21] Jacqui Rodgers, Deborah M. Riby, Emily Janes, Brenda Connolly, and Helen McConachie. 2012. Anxiety and repetitive behaviours in autism spectrum disorders and Williams syndrome: A cross-syndrome comparison. *J. Autism Dev. Disord.* 42, 2 (2012), 175–180.
- [22] Frank J. Sansosti and Kelly A. Powell-Smith. 2008. Using computer-presented social stories and video models to increase the social communication skills of children with high-functioning autism spectrum disorders. *J. Posit. Behav. Interv.* 10, 3 (2008), 162–178.
- [23] Matthew Schmidt and Noah Glaser. 2021. Investigating the usability and learner experience of a virtual reality adaptive skills intervention for adults with autism spectrum disorder. *Educ. Technol. Res. Dev.* 69, 3 (June 2021), 1665–1699.
- [24] Sarah J. Schubert, Christopher W. Lee, and Peter D. Drummond. 2011. The efficacy and psychophysiological correlates of dual-attention tasks in eye movement desensitization and reprocessing (EMDR). *J. Anxiety Disord.* 25, 1 (2011), 1–11.
- [25] Leigh Sepeta, Naotsugu Tsuchiya, Mari S. Davies, Marian Sigman, Susan Y. Bookheimer, and Mirella Dapretto. 2012. Abnormal social reward processing in autism as indexed by pupillary responses to happy faces. *J. Neurodev. Disord.* 4, 1 (2012), 1–9.
- [26] Rafael Silva, Duarte Duque, Miguel Melo, and Joao Martinho Moura. 2021. The Benefits of Virtual Reality Technology for Rehabilitation of Children with

- Autism: A Systematic Review. (December 2021), 1–8. DOI:<https://doi.org/10.1109/ICGI54032.2021.9655278>
- [27] Vítor Simões-Silva, António Marques, Catarina Pinho, Diana Gonçalves Pereira, Joana Ferreira Oliveira, and Maria Luísa Barros. 2022. Gamification Applied to Autism Spectrum Disorder. *Digit. Ther. Psychosoc. Rehabil. Ment. Heal.* (January 2022), 163–186.
- [28] Anne Snow. 2021. Social Communication Questionnaire. In *Encyclopedia of Autism Spectrum Disorders*. 4427–4429.
- [29] Mohammad Soleymani, Jeroen Lichtenauer, Thierry Pun, and Maja Pantic. 2012. A multimodal database for affect recognition and implicit tagging. *IEEE Trans. Affect. Comput.* 3, 1 (2012), 42–55. DOI:<https://doi.org/10.1109/T-AFFC.2011.25>
- [30] Mirko Uljarevic and Antonia Hamilton. 2013. Recognition of emotions in autism: A formal meta-analysis. *J. Autism Dev. Disord.* 43, 7 (2013), 1517–1526.
- [31] Andrius Vabalas and Megan Freeth. 2016. Brief Report: Patterns of Eye Movements in Face to Face Conversation are Associated with Autistic Traits: Evidence from a Student Sample. *J. Autism Dev. Disord.* 46, 1 (2016), 305–314. DOI:<https://doi.org/10.1007/s10803-015-2546-y>
- [32] Maryam Zahabi and Ashiq Mohammed Abdul Razak. 2020. Adaptive virtual reality-based training: a systematic literature review and framework. *Virtual Real.* 24, 4 (2020), 725–752. DOI:<https://doi.org/10.1007/s10055-020-00434-w>
- [33] Kun Zhang, Yishuang Yuan, Jingying Chen, Guangshuai Wang, Qian Chen, and Meijuan Luo. 2022. Eye Tracking Research on the Influence of Spatial Frequency and Inversion Effect on Facial Expression Processing in Children with Autism Spectrum Disorder. *Brain Sci.* 12, 2 (February 2022), 283. DOI:<https://doi.org/10.3390/BRAINS12020283>
- [34] Autism and Social Skills Development | Autism Speaks. Retrieved June 7, 2021 from <https://www.autismspeaks.org/tool-kit-excerpt/autism-and-social-skills-development>